

Lesson 28: One-Way Multivariate Analysis of Variance

Lesson 28 Describe how the one-way MANOVA assesses the link between a single between-subjects factor and multiple dependent variables.

Multivariate analysis of variance (MANOVA) is a multivariate extension of analysis of variance. As with ANOVA, the independent variables for a MANOVA are factors, and each factor has two or more levels. Unlike ANOVA, MANOVA includes multiple dependent variables rather than a single dependent variable. MANOVA evaluates whether the population means on a set of dependent variables vary across levels of a factor or factors. Here, we will discuss only a MANOVA with a single factor, that is, a one-way MANOVA. Each case in the SPSS data file for a one-way MANOVA contains a factor distinguishing participants into groups and two or more quantitative dependent variables.

28.1: Applications of One-Way MANOVA

One-way MANOVAs can analyze data from different types of studies.

- Experimental studies
- Quasi-experimental studies
- Field studies

We give examples of an experimental study and a quasi-experimental study.

AN EXPERIMENTAL STUDY

Dan wished to examine the effects of various study strategies on learning. Thirty students from different sections of an introductory psychology course were randomly assigned to one of three study conditions: the think condition, the write condition, or the talk condition. Students attended one general lecture after which they were placed into study rooms based on their assigned condition. Students in all rooms received the same set of study questions, but each room received different instructions about how to study. The write group was instructed to write responses to each question, the think group was instructed to think about answers to the questions, and the talk group was instructed to develop a talk that they could deliver centering on the answers to the questions. At the completion of the study session, all students took a quiz consisting of recall, application, analysis, and synthesis questions. Dan's SPSS data file includes 30 cases and five variables. The

variables include a factor differentiating the three types of study groups (write, think, and talk groups) and the four dependent variables, scores on the recall, application, analysis, and synthesis questions.

A QUASI-EXPERIMENTAL STUDY

Wanda wanted to determine whether assurances of confidentiality or anonymity have an effect on how students respond to personality and psychopathology questionnaires. She had previously administered the same three questionnaires—the Beck Depression Inventory (BDI), the 10 clinical scales of the Minnesota Multiphasic Personality Inventory-2 (MMPI-2), and the Eating Disorders Scale (EDS)—to 479 students at the beginning of three different studies. In one study, she told the students that they would complete the questionnaires anonymously. In a second study, she told the students that the results would be confidential, but she requested that they put their names on their questionnaires. In the third study, she told the students that the results would be confidential, but that she would contact them for possible referral to mental health counselors if their results warranted it. Wanda's SPSS data file has 479 cases and 13 variables, a factor differentiating among the instructional sets for the three studies (anonymity, confidentiality, and confidentiality with referral) and the 12 dependent variables (the BDI, the 10 MMPI-2 scales, and the EDS).

28.2: Understanding One-Way MANOVA

A one-way MANOVA tests the hypothesis that the population means for the dependent variables are the same for all levels of a factor, that is, across all groups. If the population means of the dependent variables are equal for all groups, the population means for any linear combination of these dependent variables are also equal for all groups. Consequently, a one-way MANOVA evaluates a hypothesis that includes not only equality among group means on the dependent variables, but also equality among group means on linear combinations of these dependent variables.

SPSS reports a number of statistics to evaluate the MANOVA hypothesis, labeled Wilks's lambda, Pillai's trace, Hotelling's trace, and Roy's largest root. (Please note that the APA style manual prefers Wilks's lambda to Wilks' lambda.) Each statistic evaluates a multivariate hypothesis that the population means on the multiple dependent variables are equal across groups. We will use Wilks's lambda because it is frequently reported in the social science literature. Pillai's trace is a reasonable alternative to Wilks's lambda.

If the one-way MANOVA is significant, follow-up analyses can assess whether there are differences among groups on the population means for certain dependent variables and for particular linear combinations of dependent variables. A popular follow-up approach is to conduct multiple

ANOVAs, one for each dependent variable, and to control for Type I error across these multiple tests by using one of the Bonferroni approaches. If any of these ANOVAs yield significance and the factor contains more than two levels, additional follow-up tests are performed. These tests typically involve post hoc pairwise comparisons among levels of the factor, although they may involve more complex comparisons. We will illustrate follow-up tests using this strategy.

The strategy that we have described (i.e., use of MANOVA, multiple ANOVAs, and pairwise comparisons) has been criticized on a number of grounds, although it is frequently used in practice. First, MANOVA is not necessary to control for Type I errors across the multiple ANOVAs and from this perspective could be skipped in applying this strategy. Second, the decision to use MANOVA and multiple ANOVAs in a sequential process is inconsistent. More specifically, conducting follow-up ANOVAs ignores the fact that the MANOVA hypothesis includes subhypotheses about linear combinations of dependent variables. Of course, if we have particular linear combinations of variables of interest, we can evaluate these linear combinations by using ANOVA in addition to, or in place of, the ANOVAs conducted on the individual dependent variables. For example, if two of the dependent variables for a MANOVA measure the same construct of introversion, then we may wish to represent them by transforming the variables to z scores, adding them together, and evaluating the resulting combined scores by using ANOVA. This ANOVA could be performed in addition to the ANOVAs on the remaining dependent variables.

If we have no clue as to what linear combinations of dependent variables to evaluate, we may choose to conduct follow-up analyses to a significant MANOVA with the use of discriminant analysis. Discriminant analysis (see Lesson 35) yields one or more uncorrelated linear combinations of dependent variables that maximize differences among the groups. These linear combinations are empirically determined and may not be interpretable.

ASSUMPTIONS UNDERLYING ONE-WAY MANOVA

Assumption 1: The Dependent Variables Are Multivariately Normally Distributed for Each Population, with the Different Populations Being Defined by the Levels of the Factor If the dependent variables are multivariately normally distributed, each variable is normally distributed, ignoring the other variables, and each variable is normally distributed at every combination of values of the other variables. It is difficult to imagine that this assumption could be met. To the extent that population distributions are not multivariate normal and sample sizes are small, the *p* values may be invalid. In addition, the power of this test may be reduced considerably if the population distributions are not multivariate normal and, more specifically, thick-tailed or heavily skewed.

Assumption 2: The Population Variances and Covariances among the Dependent Variables Are the Same across All Levels of the Factor To the extent that the sample sizes are disparate and the variances and covariances are unequal, MANOVA yields invalid results. SPSS allows us to test the assumption of homogeneity of the variance-covariance matrices with Box's *M* statistic. The *F* test from Box's *M* statistics should be interpreted cautiously in that a significant result may be due to violation of the multivariate normality assumption for the Box's *M* test, and a nonsignificant result may be due to a lack of power.

Assumption 3: The Participants Are Randomly Sampled, and the Score on a Variable for Any One Participant Is Independent from the Scores on This Variable for All Other Participants MANOVA should not be conducted if the independence assumption is violated.

EFFECT SIZE STATISTICS FOR A ONE-WAY MANOVA

The multivariate General Linear Model procedure computes a multivariate effect size index. The multivariate effect size associated with Wilks's lambda (Λ) is the multivariate eta square:

$$\text{Multivariate } \eta^2 = 1 - \Lambda^{\frac{1}{s}}$$

Here, *s* is equal to the number of levels of the factor minus 1 or the number of dependent variables, whichever is smaller. This statistic should be interpreted similar to a univariate eta square and ranges in value from 0 to 1. A 0 indicates no relationship between the factor and the dependent variables, while a 1 indicates the strongest possible relationship. It is unclear what should be considered a small, medium, and large effect size for this statistic.

28.3: The Data Set

The data set used to illustrate a one-way MANOVA is named *Lesson 28 Data File 1*. It represents data from the experimental study described earlier in the lesson. That research study involved evaluating differences in learning among three study strategies. It included four dependent variables, while here we include only two dependent variables (scores on recall and application questions). The variables in the data set are shown in Table 28.1.

Table 28.1.
Variables in Lesson 28 Data File 1

Variable	Definition
group	The factor in this study is group. The factor has three levels: think (= 1), write (= 2), and talk (= 3).
recall	Recall is the total correct score on recall questions and ranges from 0 to 8.
Applicat	Applicat is the total correct score on application questions and ranges from 0 to 8.

28.4: The Research Question

We could ask the research question to reflect mean differences or relationships between variables.

1. Mean differences: Are the population means for the scores on the recall and application tests (and linear combinations of these test scores) different for students in the three study groups?
2. Relationships between variables: Is there a relationship between the type of study strategy used and performance on the recall and application tests?

28.5: Conducting a One-Way MANOVA

Here, we will outline steps to conduct a one-way MANOVA, the univariate ANOVAs following a significant overall test, and the post hoc pairwise comparisons. Although these tests are typically sequential, we will illustrate how to conduct all three tests in a single step.

To conduct a one-way MANOVA, follow-up univariate tests, and post hoc comparisons, follow these steps:

1. Click **Analyze**, click **General Linear Model**, and then click **Multivariate**. You'll see the Multivariate dialog box shown in Figure 28.1.

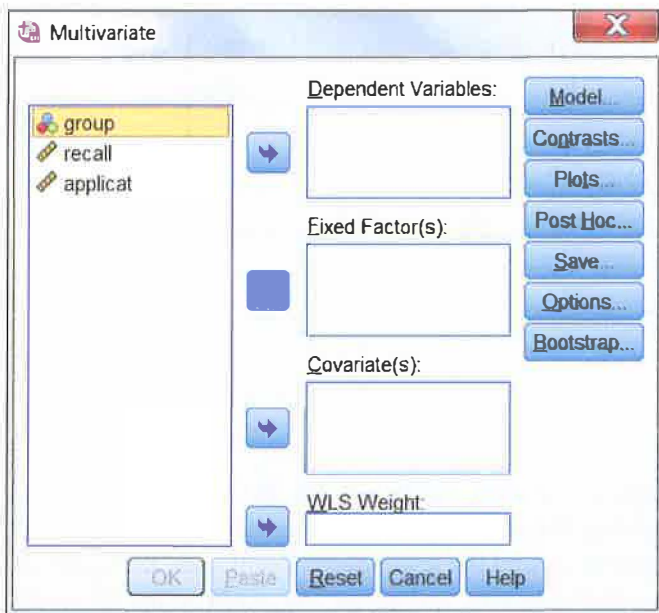


Figure 28.1. The Multivariate dialog box.

2. Click **recall**, hold down the Ctrl key and click **applicat**, and then click to move the two variables to the Dependent Variables box.
3. Click **group**, and then click to move the variable to the Fixed Factor(s) box.
4. Click **Options**. You'll see the Multivariate: Options dialog box shown in Figure 28.2.

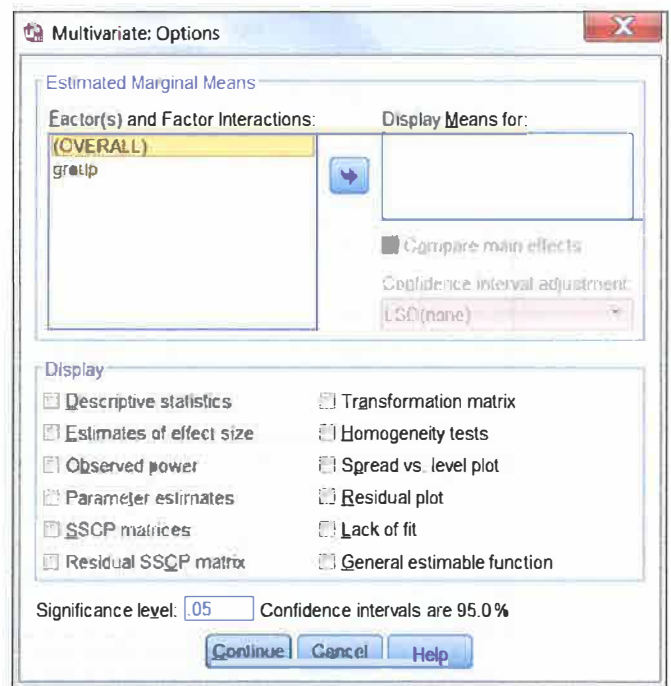


Figure 28.2. The Multivariate: Options dialog box.

5. Click **group** in the Factor(s) and Factor Interactions box, and then click to make it appear in the Display Means for box.
6. Click **Descriptive statistics**, **Estimates of effect size**, and **Homogeneity tests** in the Display box.
7. Change the Significance Level from .05 to .025. (.025 was chosen by dividing .05 by the number of dependent variables. We will talk about choosing the significance level when we discuss output for pairwise comparisons.)
8. Click **Continue**.
9. Click **Post Hoc**.
10. Under the Factors box, click **group**, and click to make it appear in the Post Hoc Tests for box.
11. In the Equal Variances Assumed box, click **Bonferroni**. Some of the other post hoc tests assuming equal variances are equally valid. (The LSD would be appropriate for our example because there are three levels of the factor, but would be inappropriate if the factor had more than three levels.)
12. In the Equal Variances Not Assumed box, click **Dunnnett's C**. Other post hoc tests that do not assume equal variances are as valid.
13. Click **Continue**.
14. Click **OK**.

SELECTED SPSS OUTPUT FOR MANOVA The results of the overall MANOVA are shown in Figure 28.3 (on page 164). The multivariate test for homogeneity of dispersion

Descriptive Statistics

Study Strategy Groups		Mean	Std. Deviation	N
Recall	Think	3.30	.675	10
	Write	5.80	1.033	10
	Talk	4.20	1.135	10
	Total	4.43	1.406	30
Application	Think	3.20	1.229	10
	Write	5.00	1.764	10
	Talk	4.40	1.174	10
	Total	4.20	1.562	30

Box's Test of Equality of Covariance Matrices^a

Box's M	6.980
F	1.039
df1	6
df2	18168.923
Sig.	.398

Tests the null hypothesis that the observed covariance matrices of the dependent variables are equal across groups.

a. Design: Intercept + group

Multivariate Tests^a

Effect		Value	F	Hypothesis df	Error df	Sig.	Partial Eta Squared
Intercept	Pillai's Trace	.962	326.035 ^b	2.000	26.000	.000	.962
	Wilks' Lambda	.038	326.035 ^b	2.000	26.000	.000	.962
	Hotelling's Trace	25.080	326.035 ^b	2.000	26.000	.000	.962
	Roy's Largest Root	25.080	326.035 ^b	2.000	26.000	.000	.962
group	Pillai's Trace	.602	5.811	4.000	54.000	.001	.301
	Wilks' Lambda	.421	7.028 ^b	4.000	52.000	.000	.351
	Hotelling's Trace	1.318	8.240	4.000	50.000	.000	.397
	Roy's Largest Root	1.275	17.215 ^c	2.000	27.000	.000	.560

a. Design: Intercept + group

b. Exact statistic

c. The statistic is an upper bound on F that yields a lower bound on the significance level.

Figure 28.3. The results of the one-way MANOVA.

matrices, Box's Test, evaluates whether the variances and covariance among the dependent variables are the same for all levels of a factor. If the F test for Box's test is significant, the homogeneity hypothesis is rejected, and we may conclude that there are differences in the matrices. The results of Box's test should be interpreted cautiously, however, in that a significant result may be due to violation of the multivariate normality assumption for this test and a nonsignificant result may be due to small sample size and a lack of power. In this example, the test for homogeneity of dispersion matrices is nonsignificant, $F(6, 18169) = 1.04, p = .40$.

Also reported on the output and of primary concern are the results of the MANOVA test. In this example, the

Wilks's Λ of .42 is significant, $F(4, 52) = 7.03, p < .01$, indicating that we can reject the hypothesis that the population means on the dependent variables are the same for the three teaching strategies. The multivariate $\eta^2 = .35$ indicates 35% of multivariate variance of the dependent variables is associated with the group factor.

SELECTED SPSS OUTPUT FOR THE UNIVARIATE ANOVAs The results of the univariate ANOVAs are shown in Figure 28.4 (on page 165).

If there are no missing data, the ANOVAs that are printed as part of the MANOVA output are identical to those obtained by conducting individual one-way ANOVAs. However, if data are missing, the univariate

Tests of Between-Subjects Effects

Source	Dependent Variable	Type III Sum of Squares	df	Mean Square	F	Sig.	Partial Eta Squared
Corrected Model	Recall	32.067 ^a	2	16.033	17.111	.000	.559
	Application	16.800 ^b	2	8.400	4.200	.026	.237
Intercept	Recall	589.633	1	589.633	629.253	.000	.959
	Application	529.200	1	529.200	264.600	.000	.907
group	Recall	32.067	2	16.033	17.111	.000	.559
	Application	16.800	2	8.400	4.200	.026	.237
Error	Recall	25.300	27	.937			
	Application	54.000	27	2.000			
Total	Recall	647.000	30				
	Application	600.000	30				
Corrected Total	Recall	57.367	29				
	Application	70.800	29				

a. R Squared = .559 (Adjusted R Squared = .526)

b. R Squared = .237 (Adjusted R Squared = .181)

Figure 28.4. The results of the univariate ANOVAs.

ANOVAs on the MANOVA output will differ from the univariate ANOVAs obtained by conducting separate ANOVAs using the Univariate General Linear Model procedure. The MANOVA procedure excludes data on all dependent variables for an individual if a score is missing on any one dependent variable. By contrast, data are excluded by the Univariate General Linear Model procedure only if a score is missing on the particular dependent variable being analyzed. To be consistent with the MANOVA results, follow-up analyses of variance should be reported for individuals who have scores on all of the dependent variables. These are the ANOVAs that are produced by the Multivariate General Linear Model procedure.

The p values for the ANOVAs on the MANOVA output do not take into account that multiple ANOVAs have been conducted. Methods may be used to control for Type I error across the multiple ANOVAs. For our application, we chose to control for Type I error using a traditional Bonferroni procedure and test each ANOVA at the .025 level (.05 divided by the number of ANOVAs conducted). Consequently, the univariate ANOVA for the recall scores was significant, $F(2, 27) = 17.11, p < .01$, while the univariate ANOVA for the application scores was nonsignificant, $F(2, 27) = 4.20, p = .026$. The ANOVA for the application scores was nonsignificant because .026 exceeded the required level of .025.

SELECTED SPSS OUTPUT FOR FOLLOW-UP PAIRWISE COMPARISONS The results of the pairwise comparisons are shown in Figure 28.5 (on page 166). We had previously controlled for Type I error across

the two univariate ANOVAs by testing each at the .025 level. To be consistent with this decision, we also need to control the probability of committing one or more Type I errors across the multiple pairwise comparisons for a dependent variable at the .025 level. We are able to maintain this familywise error rate across comparisons for a dependent variable by selecting .025 for the Significance Level in the Multivariate: Options dialog box.

TIP

The LSD procedure is a powerful method to control for Type I errors across all pairwise comparisons if a factor has three levels. If it has more than three levels, it is problematic. See Appendix B for details about the LSD procedure.

Because the ANOVA for the application scores was nonsignificant, we examined only the pairwise comparisons for the recall scores. Although more powerful methods are available, the Bonferroni approach was used to control for Type I error across the pairwise comparisons for recall. With the Bonferroni method, each comparison is tested at the alpha level for the ANOVA divided by the number of comparisons; for our example, $.025/3 = .008$. Two of the three comparisons were significant: the comparison between the thinking and writing groups and the comparison between the writing and talking groups. It should be noted that the same two comparisons were significant with Dunnett's C method.

Multiple Comparisons								
Dependent Variable		(I) Study Strategy Groups	(J) Study Strategy Groups	Mean Difference (I-J)	Std. Error	Sig.	97.5% Confidence Interval	
							Lower Bound	Upper Bound
Recall	Bonferroni	Think	Write	-2.50*	.433	.000	-3.73	-1.27
			Talk	-.90	.433	.142	-2.13	.33
		Write	Think	2.50*	.433	.000	1.27	3.73
			Talk	1.60*	.433	.003	.37	2.83
		Talk	Think	.90	.433	.142	-.33	2.13
			Write	-1.60*	.433	.003	-2.83	-.37
	Dunnett C	Think	Write	-2.50*	.380		-3.76	-1.24
			Talk	-.90	.418		-2.25	.45
		Write	Think	2.50*	.380		1.24	3.76
			Talk	1.60*	.485		.03	3.17
		Talk	Think	.90	.418		-.45	2.25
			Write	-1.60*	.485		-3.17	-.03
Application	Bonferroni	Think	Write	-1.80	.632	.025	-3.60	.00
			Talk	-1.20	.632	.206	-3.00	.60
		Write	Think	1.80	.632	.025	.00	3.60
			Talk	.60	.632	1.000	-1.20	2.40
		Talk	Think	1.20	.632	.206	-.60	3.00
			Write	-.60	.632	1.000	-2.40	1.20
	Dunnett C	Think	Write	-1.80	.680		-4.00	.40
			Talk	-1.20	.537		-2.94	.54
		Write	Think	1.80	.680		-.40	4.00
			Talk	.60	.670		-1.57	2.77
		Talk	Think	1.20	.537		-.54	2.94
			Write	-.60	.670		-2.77	1.57

Based on observed means.

The error term is Mean Square(Error) = 2.000.

*. The mean difference is significant at the .025 level.

Figure 28.5. The results of the post hoc comparison procedures.

28.6: Using SPSS Graphs to Display the Results

Graphical methods, such as boxplots, can be used in the Results section to allow the reader to evaluate the differences among groups. The boxplots that display the distributions for the multiple dependent variables of a MANOVA are created using a method that is slightly different from the ones used for an independent-samples *t* test or one-way ANOVA. To create the boxplot shown in Figure 28.6 (on page 167), conduct the following steps:

1. Click **Graphs**, click **Legacy Dialogs**, and then click **Boxplot**.
2. Click **Clustered** and **Summaries for Separate Variables** in the Boxplot dialog box.
3. Click **Define**.
4. Click **group**, and then click  to move it to the Category Axis box.
5. Holding down the Ctrl key, click **recall** and **applicat**, and then click  to move them to the Boxes Represent box.
6. Click **OK**. Make appropriate edits to obtain Figure 28.6.

28.7: An APA Results Section

A one-way multivariate analysis of variance (MANOVA) was conducted to determine the effect of the three types of study strategies (thinking, writing, and talking) on the two dependent variables, the recall and the application test scores. Significant differences were found among the three study strategies on the dependent measures, Wilks's $\Lambda = .42$, $F(4, 52) = 7.03$, $p < .01$. The multivariate η^2 based on Wilks's Λ was quite strong, .35. Table 28.2 contains the means and the standard deviations on the dependent variables for the three groups.

Table 28.2.

Means and Standard Deviations on the Dependent Variables for the Three Groups

Study strategy	Recall		Application	
	M	SD	M	SD
Thinking	3.30	0.67	3.20	1.23
Writing	5.80	1.03	5.00	1.76
Talking	4.20	1.14	4.40	1.17

Analyses of variances (ANOVAs) on the dependent variables were conducted as follow-up tests to the

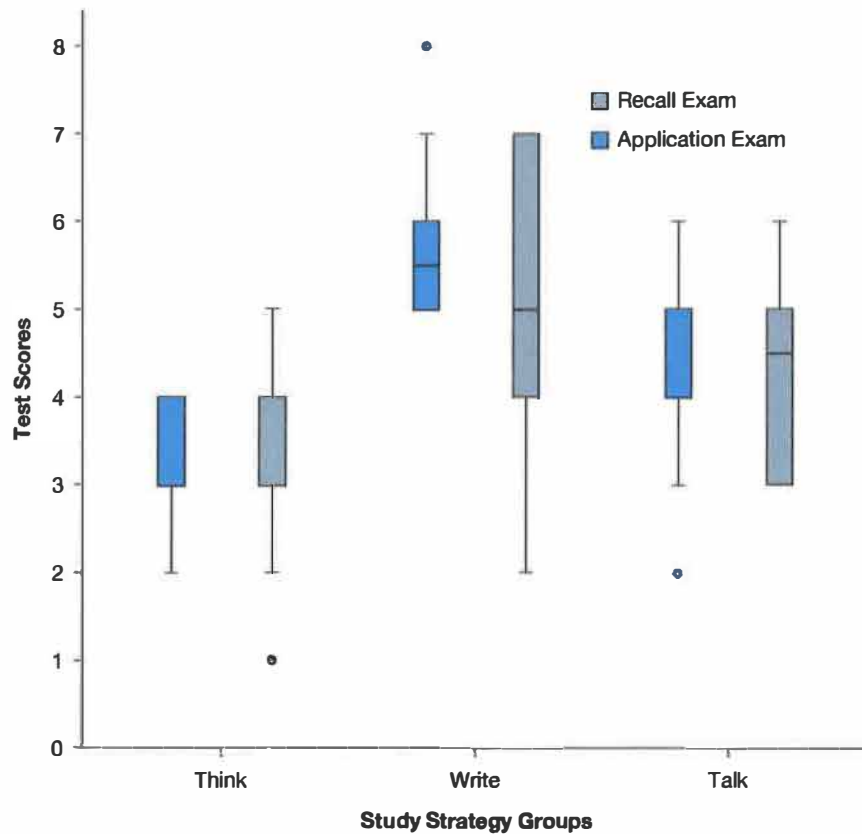


Figure 28.6. Distributions of application and recall scores for the three study strategy groups.

MANOVA. Using the Bonferroni method, each ANOVA was tested at the .025 level. The ANOVA on the recall scores was significant, $F(2, 27) = 17.11, p < .01, \eta^2 = .56$, whereas the ANOVA on the application scores was nonsignificant, $F(2, 27) = 4.20, p = .026, \eta^2 = .24$.

Post hoc analyses to the univariate ANOVA for the recall scores consisted of conducting pairwise comparisons

to find which study strategy affected performance most strongly. Each pairwise comparison was tested at the .025 divided by 3 or .008 level. The writing group produced significantly superior performance on the recall questions in comparison with either of the other two groups (see Table 28.2). The thinking and talking groups were not significantly different from each other.

Exercises: Lesson 28

The data for Exercises 1 through 4 are in the data set named *Lesson 28 Exercise File 1*. The data are from the following research problem.

Dana wished to evaluate the effects of conflict-resolution training on the social skills of elementary school children. Fourth-grade students in a school district were randomly assigned to one of three groups. Two groups received training in conflict resolution from either a cognitive-based approach (group = 1) or a behavior-based approach (group = 2). The third group did not receive formal instruction in conflict resolution (group = 3). To measure the effects of the treatment,

a social problem-solving inventory (SPSI) was administered to the children, and rating scales were completed by the teachers (rate_t) and by the parents (rate_p).

1. Conduct a MANOVA to evaluate the effects of the training programs. Identify the following values on the output:
 - a. Wilks's lambda
 - b. F statistic associated with Wilks's lambda
 - c. Multivariate η^2 associated with Wilks's lambda
 - d. Univariate F test for the teacher rating
 - e. Mean parent rating for group 2

2. Conduct the appropriate follow-up tests.
3. Write a Results section reporting the results of her analyses.
4. Create a boxplot to display graphically Dana's results.

The data for Exercises 5 and 6 are in the data set named *Lesson 28 Exercise File 2*. The data are from the following research problem.

Vicki was interested in whether fathers of children with disabilities feel competent in parenting their children. To address this question, she found 60 fathers who fall into one of three categories: (1) fathers of children with no disabilities, (2) fathers of children with physical disabilities, and (3) fathers of children with cognitive impairments. She administered the

Perceived Parenting Competence Scale (PPCS), a self-report measure. It has three subscales: (1) instrumental competence, which assesses caretaking responsibilities (instrum); (2) emotional parenting competence, which evaluates the quality of emotional support provided to the child (emot); and (3) play competence, which measures the quality of the recreational time spent with the child (play). Vicki's SPSS data file contains 60 cases and four variables, a factor indicating the disability status of the child and scores on the three PPCS subscales.

5. Conduct a one-way MANOVA to evaluate differences among the three types of fathers on the three perceived competence variables. Is the multivariate test significant?
6. Write a Results section.

Lesson 29: One-Way Repeated-Measures Analysis of Variance

Lesson 29 Outline how the one-way repeated-measures ANOVA assesses the mean differences between multiple levels of a factor.

With one-way repeated-measures designs, each subject or case in a study is exposed to all levels of a qualitative variable and measured on a quantitative variable during each exposure. The qualitative variable is referred to as a repeated-measures factor or a within-subjects factor. The quantitative variable is called the dependent variable.

To conduct a repeated-measures ANOVA in SPSS, we do not specify the repeated-measures factor and the dependent variable in the SPSS data file. Instead, the SPSS data file contains several quantitative variables. The number of quantitative variables is equal to the number of levels of the within-subjects factor. The scores on any one of these quantitative variables are the scores on the dependent variable for a single level of the within-subjects factor.

Although we do not define the within-subjects factor in the SPSS data file, we specify it in the dialog box for the General Linear Model Repeated-Measures procedure. To define the factor, we give a name to the within-subjects factor, specify the number of levels of this factor, and indicate the quantitative variables in the data set associated with the levels of the within-subjects factor.

29.1: Applications of One-Way Repeated-Measures ANOVA

One-way within-subjects ANOVAs can analyze data from different types of studies.

- Experimental studies
- Quasi-experimental studies
- Field studies
- Longitudinal studies

We illustrate a longitudinal study and an experimental study.

LONGITUDINAL STUDY

Janice is interested in determining whether the average man wants to express his worries to his wife more (or less) the longer they are married. Janice develops the Desire to Express Worry (DEW) scale. She has men take the DEW when they initially get married and then at their 5th, 10th, and 15th wedding anniversaries. She decides to conduct a one-way within-subjects ANOVA that involves only those 30 men who made it to their 15th wedding anniversary. The within-subjects factor is time with four levels (0, 5, 10, and 15 years of marriage), and the dependent variable is the DEW scores at the four times. Janice's SPSS data file has 30 cases and four variables. The four variables are the initial DEW score and the DEW scores after 5, 10, and 15 years of marriage.

EXPERIMENTAL STUDY

Shep is interested in determining if teachers feel greater stress when coping with problems associated with students, parents, or administrators. To make this